INDEPENDENT QUESTION SELECTION

Making a three-dimensional passive quantum memory efficiently writable

A decoder-history construction for thermal initialization

5 November 2026

Selected full question. Can a uniform, polynomial-cost procedure write an arbitrary unknown qubit, including its entanglement with a reference, into a three-dimensional local passive quantum memory from product ancillary states, then recover it after $\exp(cN^{\eta})$ holding time at fixed positive temperature, with an explicit preparation, control, uncertainty and reuse bill?

Target family. Use the random-embedding family whose lifetime is proved in Balasubramanian–Davydova–Lin (BDL), arXiv:2605.10943v1. Their separately proposed explicit embedding is not substituted for the proved family. The external lifetime theorem is a cited input, not a proof independently reproduced here.[1]

Constructive opening. Keep the whole multiscale decoder history as local classical variables. If each syndrome has one valid history and each connected nonzero history component has energy proportional to its size, a fully specified low-temperature polymer sampler prepares the needed syndrome law. A decoder-frame Pauli then loads the unknown qubit without measuring its logical information.

Completed in this round. An exact total-variation/diamond-norm loading identity; a conditional history-to-sampler theorem with an explicit mixing bound; a source-based BDL instantiation checked against the stated local lemmas; exact finite examples; and a comparison against sector mixing, Code Swendsen–Wang, regular matroids, and recent decoding-based Gibbs methods.

Still open. The novelty comparison: Code Swendsen–Wang or regular-matroid sampling may already supply the same operation. Neither possibility has been ruled out. Two October 2026 decoding papers were available only through primary abstracts. The argument below takes BDL's stated lemmas as inputs; it does not independently validate the complete external proof, supply an implementation of that enormous family, or establish robust hardware operation.

Selection status. This is one nominated research question and a proved conditional opening. It is not a claim that initialization, practical quantum hardware, or a civilisation-changing discovery has already been completed.

1 The consequence that the full question earns

Consider a known commuting stabilizer Hamiltonian on N qubits,

$$H_N = \sum_{a=1}^{m} J_a \frac{I - S_a}{2}, \qquad m = O(N), \quad 0 < J_{\min} \le J_a \le J_{\max}. \tag{1}$$

The coefficient convention assigns energy J_a to a violated check. The primary source target uses $J_a = 1$; Section 4 derives a uniform extension to positive inhomogeneous coefficients from the same component construction. The stabilizers have bounded support and bounded geometric range in three dimensions. Access to these interactions is part of the physical model.

The selected operation is *write, hold, read*. The write and read procedures may use active gates, resets, measurements, and classical computation. During the holding interval there are no scheduled syndrome measurements, corrections, or refresh gates. The thermal bath and static Hamiltonian remain present. Requiring the write procedure itself to be passive would add a restriction that the useful operation does not need. A polynomial ground-state encoder with a proved comparable ground-start lifetime would also solve the question; thermal initialization is the constructive route developed here, not a mandatory intermediate state of every solution.

BDL supply the metastability input for their decoder-defined thermal encoding. Their initialization discussion leaves efficient passive preparation open; their existing sector-mixing comparison needs a local metacheck hypothesis absent from their construction. We are seeking a broader active initialization procedure. Their actual proof concerns the random-embedding family, and does not establish the lifetime of the separate explicit-embedding proposal.[1]

For the proposed initialization, the desired operational statement is

$$\frac{1}{2}\left\| \widetilde{\mathcal{D}}_N \, \Phi_{N,t} \, \widetilde{\mathcal{E}}_N - \mathrm{id} \right\|_\diamond \le \delta_{\mathrm{load}} + \delta_{\mathrm{control}} + \delta_{\mathrm{read}} + C \Lambda_N t \, e^{-cN^\eta}. \tag{2}$$

Here Λ_N records the bath-rate normalization; for bounded local rates it is polynomial in N. The theorem must specify the Lindbladian class and its rates, rather than hiding an arbitrarily strong bath inside the time unit. The full state is compared with an arbitrary reference, so the statement stores quantum information rather than only a classical basis label.

A concrete sufficient noise model is a Davies-type generator built from bounded one-site Pauli system–bath couplings, resolved into their local energy-transition operators, with rates satisfying thermal detailed balance and a fixed per-site rate bound Γ. The energy-resolved jumps are not simply bare Pauli operators with state-independent rates. The general stability criterion transfers protection of the coupling operators to their energy-resolved jumps and bounds $\Lambda_N = O(\Gamma N)$.[10] The model treats the Markovian semigroup as the holding dynamics; a finite-coupling bath approximation uniform over exponential times would be an additional physical theorem.

At fixed overall error, (2) would allow a holding time $t = e^{c' N^\eta}$ with $c' < c$, while write and read costs remain polynomial in N and the desired inverse accuracy. This would establish *efficient accessibility* of a protected three-dimensional thermal quantum state. It would not establish a practical device, constant-noise fault tolerance during writing, or robustness to general coherent Hamiltonian errors.

1.1 Why constant loading error does not grow with the storage time

Suppose the ideal encoder is $\mathcal{E}_\beta$, and an implemented one obeys $\frac{1}{2}\left\|\widetilde{\mathcal{E}} - \mathcal{E}_\beta\right\|_\diamond \le \delta$. For every holding channel Φ_t and ideal readout channel $\mathcal{D}$,

$$\frac{1}{2}\left\| \mathcal{D}\Phi_t\widetilde{\mathcal{E}} - \mathcal{D}\Phi_t\mathcal{E}_\beta \right\|_\diamond \le \delta. \tag{3}$$

This is contractivity under channels. The initialization discrepancy is paid once; it is not a fresh error injected at every instant. Therefore polynomial-accuracy preparation can support exponentially long storage in the stated bath model. The same argument does *not* remove errors continuously injected by an unmodelled bath or unwanted coherent evolution.

The primary lifetime input is stated for each one-qubit density matrix. This suffices to obtain a reference-preserving version with a dimension-dependent constant. If a Hermiticity-preserving map A obeys $\|A(\rho)\|_1 \le r$ for every one-qubit state, write a pure reference-system input in Schmidt form. The diagonal contributions have total norm at most r. Since $|0\rangle\langle 1| = (X + iY)/2$ and $\|A(X)\|_1, \|A(Y)\|_1 \le 2r$, the off-diagonal contributions total at most $2r$. Thus $\|A\|_\diamond \le 3r$ for a difference of channels on this input space. Only an absolute prefactor changes in (2).

2 The exact loading reduction

Let $\mathcal{S}$ be the valid stabilizer syndromes and V_0 a ground-space encoding isometry for one qubit. Fix the *same decoder frame* that defines the protected logical observables. For every $s \in \mathcal{S}$, let F_s be its chosen Pauli correction to the ground space, and set $V_s = F_s^\dagger V_0$. The ranges of distinct V_s are orthogonal syndrome sectors.

For a classical distribution p on $\mathcal{S}$, define

$$\mathcal{E}_p(\rho) = \sum_{s\in\mathcal{S}} p(s) V_s \rho V_s^\dagger. \tag{4}$$

Its implementation is concrete: sample s, encode the input into the ground code, and apply $F_s^\dagger$. A syndrome sample can be generated before the unknown input arrives, and it can be consumed once when loading.

Proposition 1 (Sampling error equals channel error)**.** *For any distributions p, q on the valid syndrome set,*

$$\boxed{\frac{1}{2}\|\mathcal{E}_p - \mathcal{E}_q\|_\diamond = \mathrm{TV}(p,q).} \tag{5}$$

Proof. On any state ρ_{LR}, the output difference is block diagonal in the mutually orthogonal syndrome sectors, with block $(p(s) - q(s))(V_s \otimes I_R)\rho_{LR}(V_s^\dagger \otimes I_R)$. Each positive state in this expression has trace one. Its trace norm is therefore exactly $\sum_s |p(s) - q(s)|$, independently of the input or reference. Maximizing over reference-system states gives the result. □

The thermal choice is $p_\beta(s) \propto e^{-\beta E(s)}$. All valid syndrome sectors have the same logical dimension, so there is no additional syndrome-dependent quantum degeneracy to insert. In the CSS setting the two syndrome distributions factor, allowing two samplers with an additive accuracy budget.

If G is a useful set of syndromes, then

$$\mathrm{TV}\big(p_\beta, p_\beta(\cdot \mid G)\big) = p_\beta(G^c). \tag{6}$$

Sampling the conditional law to error ε gives loading error at most $\varepsilon + p_\beta(G^c)$. A rare excluded set need only fit the operational error budget. It need not be sampled exactly.

An initial version of the opening used this conditioning step, with an exponentially small preparation floor at fixed N. Section 4 removes that restriction: the constant-size base code can have its entire validity relation retained. Under the compiler premises the resulting sampler targets the full syndrome law, so its accuracy is independent of the rare-instability probability. The useful operation still requires only fixed or inverse-polynomial initialization error; arbitrarily accurate sampling is a stronger available output of the conditional construction.

Strong-comparator correction. An efficient sampler for the *full* quantum Gibbs state also solves the sampling step: measure its commuting stabilizers, retain the resulting classical syndrome, and load a fresh unknown input with (4). The sampler may randomize the logical sector. That is not a valid reason to reject it as a competitor. Conversely, an arbitrary syndrome representative is insufficient without tracking its possible logical Pauli relative to the protected decoder frame.

3 The constructive opening: histories as local constraints

The starting obstacle is that the physical syndrome distribution may have large, nonlocal parity relations. Independent thermal defects conditioned on those relations can make rejection exponentially expensive. A local failure-probability bound also does not give a sampler.

The proposed representation keeps the intermediate decoder syndromes instead of eliminating them. Write a history as $h = (s_\ell, s_{\ell-1}, \dots, s_0)$ with s_ℓ physical and s_0 a valid syndrome of a constant-size base code. Local correction functions are fixed deterministic functions of their inputs; tie-breaking is fixed once. The correction variables themselves need not be sampled. Their residual equations can be checked as local constraints.

Form the *constraint-primal graph*: connect any two history variables that occur in the same constraint. This graph can differ from the graph used to witness rare failures. It also differs from the three-dimensional hardware layout. Its role here is classical computation.

Theorem 1 (Conditional history-to-sampler compiler)**.** *Suppose a family of history models has the following properties, uniformly in system size.*

1. *There are $M = \mathrm{poly}(N)$ variables, each with an alphabet of size at most q, including a distinguished zero. The constraint-primal graph has maximum degree at most Δ, with fixed q, Δ.*
2. *The all-zero history is valid. Valid histories are in bijection with a chosen set G of physical syndromes. This bijection and its inverse are computable in polynomial time.*
3. *Every constraint is explicitly supplied and locally evaluable. Feasibility of a connected support of r variables, with prescribed nonzero labels and zero outside, is checkable in $e^{O(r)}\,\mathrm{poly}(N)$ time.*
4. *The energy is additive over physical syndrome coordinates. Every nonzero connected component γ of every valid history satisfies*

$$E(\gamma) \geq a|\gamma| \qquad (a > 0 \text{ fixed}). \tag{7}$$

5. *Activities can be computed to prescribed precision with polynomial bit cost.*

Let $\kappa = (q-1)\max(2,\Delta)^2$ and suppose

$$x := \kappa e^{-\beta a} \leq \frac{1}{8(\Delta+1)}. \tag{8}$$

Then the Gibbs distribution on G can be sampled to total-variation error ε in $\mathrm{poly}(N, 1/\varepsilon)$ time. The polynomial and temperature threshold depend on the fixed structural constants. The theorem does not assume that physical thermalization mixes rapidly.

3.1 Proof: why this is an exact polymer representation

A polymer is a nonempty connected support with nonzero labels, whose zero extension is a valid history. Distinct polymers are compatible when their supports are disjoint and nonadjacent in the primal graph.

Take a valid history and one connected component of its nonzero support. Any factor meets at most one such component: two nonzero variables in its scope would be adjacent. After deleting

the other components, a factor therefore sees either its original valid assignment or the all-zero valid assignment. Every component is consequently an individually valid polymer. Conversely, combining any compatible collection preserves every factor for the same reason.

This gives an exact bijection between histories and compatible polymer collections. Uniqueness of the history for each syndrome removes any hidden multiplicity in the physical measure. Additive energy gives weight

$$W(\Gamma) = \prod_{\gamma \in \Gamma} w(\gamma), \qquad w(\gamma) = e^{-\beta E(\gamma)} \le e^{-\beta a |\gamma|}. \tag{9}$$

This is standard positive polymer structure; low-temperature polymer sampling is established machinery.[7, 8] The useful new object to construct is the decoder-history representation with the stated properties.

3.2 Proof: explicit truncation and mixing bounds

For a fixed root vertex, there are at most κ^r labeled connected supports of size r. One elementary bound encodes a spanning-tree traversal by at most $2r-2$ neighbor choices, then assigns at most $(q-1)^r$ labels. A canonical traversal makes this an injective encoding of supports; the displayed κ is deliberately conservative.

For any polymer, its Gibbs inclusion probability is at most $w(\gamma)$: removing it maps collections containing it into compatible collections, and the full partition function includes every image. Hence

$$\Pr[\exists \gamma : \ |\gamma| > L] \le M \sum_{r > L} x^r = \frac{M x^{L+1}}{1-x}. \tag{10}$$

Choose L so this is at most $\varepsilon/2$. Thus $L = O(\log(M/\varepsilon))$. Enumerate all admissible polymers of size at most L using bounded-degree connected-set enumeration and the supplied feasibility test. Their number satisfies $K \le M \sum_{r \le L} \kappa^r$, which is polynomial in M/ε.

Run the following finite heat-bath chain on compatible collections of these K polymers. Select a polymer uniformly. If any incompatible polymer is present, set the selected polymer absent. Otherwise put it present with probability $w/(1+w)$. The empty collection is a valid starting state. Removing polymers connects every state to it, and the chain has the desired truncated Gibbs law.

For coupling, use weighted Hamming distance, giving a disagreement on polymer γ weight $|\gamma|$. Consider configurations differing only on γ. Updating γ removes the disagreement with probability $1/K$. Updating an incompatible polymer γ' creates a disagreement with probability at most $w(\gamma')/K$. Since every incompatible support meets the closed neighborhood of γ,

$$\sum_{\gamma' \not\sim \gamma} |\gamma'| w(\gamma') \le (\Delta+1)|\gamma| \sum_{r \ge 1} r x^r \tag{11}$$

$$= (\Delta+1)|\gamma| \frac{x}{(1-x)^2} \le \frac{1}{4}|\gamma|. \tag{12}$$

Thus one step contracts this distance by at least $1 - 3/(4K)$. Any two compatible collections are joined by removing their differences and then adding the new ones, so the adjacent-pair coupling extends along a shortest path. The diameter is at most $2M$ and every nonzero distance is at least one. After

$$T \ge \frac{4K}{3} \log \frac{4M}{\varepsilon} \tag{13}$$

steps the distance to the truncated law is at most $\varepsilon/2$ in total variation. Together with (10), this proves the exact-arithmetic statement. If $K = 0$, output the zero history. For finite precision, run these bounds with target $\varepsilon/2$ and reserve $\varepsilon/2$ for numerical error. Approximating each transition probability to error $\varepsilon/(2T)$ then suffices by a stepwise coupling; the needed bit precision is logarithmic in T/ε.

4 The source-based construction and remaining audit

The central target is a *syndrome-only* local compiler. At each scale use the supplied local correction functions to enforce a relation of the schematic form

$$s_i = \partial_i f_i(s_i) + F_i s_{i-1}, \qquad s_{i-1} = R_i(s_i), \qquad s_0 \in B_0. \tag{14}$$

Here B_0 is the valid-syndrome space of the fixed base code. The notation abstracts the source's maps (whose modified base is level 2); an implementation must use their exact indices, local complexes, and deterministic choices. It must retain intermediate levels rather than compose a large map and call the resulting long-range constraint local. The complete base-validity relation is one constant-arity factor, not a growing global constraint.

If the isolated valid histories obey alternating nonincrease and halving of syndrome weight, then

$$\sum_i |s_i| \le 4|s_\ell|. \tag{15}$$

Counting only syndrome bits yields $E(\gamma) \ge J_{\min}|\gamma|/4$. Correction bits may instead be included, but then their support must be charged to syndrome support with a fixed constant. A gate-by-gate transcript that costs $\log N$ active auxiliary bits per physical defect would invalidate the fixed-temperature conclusion.

The equations recover global compatibility locally. Start with a valid base syndrome $s_0 = \partial_0 e_0$. The cochain property gives $F_i \partial_{i-1} = \partial_i F_i^{(1)}$. Inductively, (14) expresses every s_i as a coboundary. The accumulated corrections, including a base correction, supply its physical Pauli representative. A zero-extended polymer satisfies the same constraints and base-validity factor, so the same argument applies to it. Once its history agrees with the actual decoder, the global contraction theorem applies to that isolated history. It is unnecessary to identify it with the source's rare-failure witness.

There is a concrete way to compile local tables even when the source decoder is defined only on valid syndromes. For every constant-size local syndrome view, solve a binary linear extension problem to determine whether it extends to a valid full syndrome. For an extendable view, evaluate the actual deterministic decoder on a canonical extension. Local computability makes the answer independent of the extension. For an unextendable view, choose a fixed default. Enforce both the output table and the residual equation in (14). Upward induction from the valid base proves that every accepted level is a coboundary, so only the genuine, extendable branches are ever used. This avoids assuming a global validity oracle inside polymer enumeration. Its preprocessing is polynomial if neighborhoods and their alphabets are fixed and the source maps are explicitly computable.

Determinism now gives a bijection onto *all* valid physical syndromes. Conversely, every physical syndrome and its actual decoder history satisfy these constraints. The all-zero history is accepted, and component deletion follows from the primal-graph argument. Thus the conditional construction has no discarded-syndrome error floor. The physical loading step retains the protected decoder frame. The auxiliary decoder used to certify the classical sampling distribution may be different: the full Gibbs syndrome law depends on the Hamiltonian, not the chosen error representative. This avoids imposing unnecessary agreement between sampling certificates and the logical frame.

The source inputs are specific: Definition 4.3 supplies local correction/coarse-output dependence; Theorems 4.6–4.7 supply the residual identities and contraction; Section 5.1.1 supplies a constant-size base. For bounded residual incidence, choose homological contractions inside each replacement block. The terminal degree-two input lies in the local Z sector, on which the inter-block perturbation vanishes. In Appendix A's inclusion formula $i' = i + h\Sigma i$, the correction then vanishes on that input. Consequently $F_i^{(2)}$ is a local representative inclusion, with bounded row and column incidence.[1]

Uniform local choices. The tables need not be nonuniform advice. For an X replacement, enumerate a local correction f_x with $\delta_x f_x = u_x$ and $\sum_{q\sim x} |g_{xq} f_x| \le |u_x|$, then a Q correction with

the analogous outgoing-weight inequality. Select the first admissible correction in a fixed ordering, with zero first; use a fixed default on inadmissible views. For a Z replacement, take the local syndrome parity and move it to the same representative point used in $F^{(2)}$. All searches are within fixed-size blocks. The source's local correction lemmas guarantee admissibility on physical inputs; the upward induction excludes spurious accepted histories arising from defaults.

Proposition 2 (Source-conditional thermal initialization)**.** *Accept the random-embedding construction, local correction and contraction lemmas of BDL, with the blockwise cochain representatives above. At a fixed sufficiently low positive temperature, its full Gibbs syndrome law admits a uniform classical sampler with total-variation error ε and cost* $\mathrm{poly}(N, 1/\varepsilon)$*. Coupled to the protected-frame loader, the stated active-control resources and the cited holding theorem, it gives the channel guarantee* (2)*.*

Proof. Choose individual syndrome bits as variables, so $q = 2$, and include the complete finite base factor. All local block sizes, dependency radii and row/column incidences are bounded independently of N; completing factor scopes to cliques therefore gives fixed degree. The residual induction proves bijection with the full physical syndrome space. Every isolated component is a valid decoder history and obeys (15), yielding $a = J_{\min}/4$. Theorem 1 applies at an inverse temperature satisfying both its threshold and the holding theorem's. Proposition 1 converts its error directly into a reference-preserving loading error. Channel contractivity and the independent write/read error budgets give (2). □

Precise status. The sparse-inclusion argument and compiler are deductions from the cited source properties, not claims quoted from the paper. Their directions, domains and local choices were independently checked within this session. The complete external proof was not independently re-proved, and no full-family implementation was run. The major unresolved comparison is whether a stronger existing sampler already supplies the same operation; a novelty or priority claim would be premature.

Potential failures are concrete: a local solver that depends on remote choices; a hidden global syndrome test; more than one accepted history per syndrome; failure of contraction after component isolation; an auxiliary support overhead growing with N; or a factor graph whose degree grows with N. Any of these would require changing the representation or the selected attack.

4.1 A uniform bound for positive commuting coupling disorder

For this consequence choose the same admissible decoder for the history certificate and protected frame, with the base correction retaining its stated small-error guarantee. Use one binary variable per syndrome bit. A nonzero syndrome at base level 2 belongs to a polymer containing a base variable. Alternate contraction between levels 2 and $2k$ forces that isolated component to have

$$|s_{2k}^{\gamma}| \geq 2^{k-1}, \qquad |\gamma| \geq 2^{k-1}. \tag{16}$$

Let B be the number of base syndrome variables, a family constant. With $x = \kappa e^{-\beta J_{\min}/4}$, inclusion probabilities and rooted counting give

$$\Pr[s_2 \neq 0] \leq B \sum_{r \geq 2^{k-1}} x^r = \frac{B\, x^{2^{k-1}}}{1-x}. \tag{17}$$

This estimate uses each actual positive coupling in the activities and only its lower bound in the inequality. It is uniform over all fixed $J_a \in [J_{\min}, J_{\max}]$. Apply it to both CSS sectors and add their probabilities. Since $N = e^{\Theta(k)}$, the result is $b_N \leq Ce^{-cN^{\eta}}$ for some fixed $\eta > 0$.

Base-zero histories give the protected subspace in the source decoder argument. That combinatorial condition is independent of coupling strengths. The dressed logical operators and the syndrome projector commute with every stabilizer, hence with the weighted Hamiltonian. The general Gibbs-stationary protected-observable theorem therefore gives the same form of storage bound for the weighted model.[10] This is a deduction from the compiled representation and protection criterion, not an assumed noncommuting perturbation theorem.

Thus bounded positive coefficient variation is compatible with the proposed write/hold/read result. Preparing with estimated coefficients still has the calibration error in Section 7. Couplings and bath parameters remain fixed during a hold; the argument does not cover arbitrary temporal drift, changes to the stabilizer operators, extra logical terms, or noncommuting perturbations.

5 A decisive finite example, and a counterexample

Consider a four-child tree. At each internal vertex with parent bit s and child bits a, b, c, d, impose

$$a \oplus b = c \oplus d = s, \tag{18}$$

and fix the root to zero. Only leaf bits cost energy. Given all leaves, the internal history is unique when it exists. Each nonzero parent forces at least two nonzero children, so the total nonzero history size is less than twice the number of defective leaves.

Eliminating internal variables creates growing-support constraints on the leaves. Keeping them gives bounded-arity local constraints. Join all variables in each factor scope. The component deletion and recombination proof then applies exactly. This is a finite demonstration of the proposed representation, not an implementation of the BDL memory and not evidence that existing graph-based samplers fail.

Let $z = e^{-\beta J}$ and let $Z_h(0), Z_h(1)$ be the depth-h partition functions at fixed root. Direct calculation gives

$$Z_{h+1}(0) = (Z_h(0)^2 + Z_h(1)^2)^2, \tag{19}$$

$$Z_{h+1}(1) = 4Z_h(0)^2 Z_h(1)^2. \tag{20}$$

Consequently $r_h = Z_h(1)/Z_h(0)$ obeys

$$r_{h+1} = \frac{4r_h^2}{(1+r_h^2)^2} \le 4r_h^2, \qquad r_h \le \frac{(4z)^{2^h}}{4}. \tag{21}$$

For $z < 1/4$ this is stretched-exponential suppression in the number 4^h of leaves. A direct recursive sampler works for this toy; no novel algorithm is claimed for it.

The accompanying exact rational check enumerates all 65,536 leaf strings at depth two. It finds 1,024 valid histories and 300 distinct polymers, verifies component deletion, 246 compatible-pair recombinations, additive energy, the support bound, and the equality of recursive sampling probabilities with the target law.

Why locality alone is insufficient. Replace the four-child rule by binary XOR, with parent $s = a \oplus b$ and root zero. Put two nonzero leaves in opposite main branches of a depth-h tree. Each branch propagates a nonzero syndrome to the root's children, and the root cancels. The history contains $2h$ nonzero variables but leaf energy only $2J$. It remains uniquely determined and locally checkable, yet the energy per history variable tends to zero. Thus the contraction condition in the proposed compiler is a substantive requirement.

6 The strongest existing methods are part of the test

The candidate is not selected on the premise that low-temperature preparation is generally hard. The following existing approaches can remove the need for this construction if their hypotheses hold for the target.

Method	What it supplies	Actual comparison for this target
Ground-state stabilizer encoding	Polynomial-size Clifford encoding from a known stabilizer description.	Supplies V_0, but not the thermal syndrome law used by the lifetime input. Ground-state stability under the bath is not obtained merely by writing down a thermal-state theorem.
Sector mixing, Bergamaschi–Gheissari–Liu[2]	Rapid preparation in logical sectors for a class of lattice memories; a 4D toric-code application.	Its local-redundancy assumptions require a separate check. The BDL source specifically identifies its metacheck condition as an obstruction to direct application.
Code Swendsen–Wang, v4[3]	Correct global updates for code Hamiltonians; rapid mixing when the check matroid is within $\Delta = O(\log N)$ of a graphic or cographic form.	Strongest concrete competing route. No BDL representation satisfying the criterion has been constructed here, and no exclusion has been proved. One encoded qubit does not bound the relation-space defect Δ.
Regular-matroid approximation, Goldberg–Jerrum[9]	A fully polynomial randomized approximation scheme for the ferromagnetic Ising partition function on regular matroids.	Deletion/contraction self-reduction gives sampling, so regularity of the target check matroids could suffice even if a graphic/cographic representation is absent. BDL regularity is not settled here.
Commuting-Hamiltonian reduction, Hwang–Jiang[4]	Reductions to classical Gibbs sampling for specified 2-local and 2D 4-local classes.	A reduction is useful, but still needs a suitable classical sampler or a proof that this 3D family falls in its tractable class.
Quantum decoding HDQI[5]	Primary abstract ties reachable temperature to decoder thresholds, with a random-sign guarantee.	A proved implication for the all-positive, sufficiently low-temperature BDL distribution has not been established here. Full-text access failed.
Relative decoding[6]	Primary abstract transfers a source TFD preparation guarantee using a quotient-code decoder.	Must supply an efficient source, the relative decoder, and a temperature range covering the target. Full-text applicability remains unresolved.
Positive polymer algorithms[7, 8]	Efficient sampling under suitable local enumeration and weight-decay conditions.	This machinery supports the proposed opening. The new work would be the faithful history compiler, not the existence of polymer sampling.

6.1 The specific Code Swendsen–Wang challenge

Let h be the binary check matrix of one CSS sector. In the current primary theorem, an admissible graphic comparison has an incidence matrix g on the same check-coordinate set with

$$\ker(h^T) \subseteq \ker(g^T), \qquad \dim\ker(g^T) - \dim\ker(h^T) = \Delta. \tag{22}$$

There is a dual condition for the syndrome space. The mixing dependence is $2^{O(\Delta)}\,\mathrm{poly}(N)$. Nonlocal metachecks do not rule this condition out, because the comparison graph need not be geometrically local. Nor does preserving one logical qubit imply small Δ: middle cohomology and the spaces of check relations are different objects.[3]

Therefore a construction of such a graph for both BDL sectors would defeat the claimed need for our polymer route. It would also furnish a simpler solution to the loading task via Proposition 1. This possibility is retained as a live challenge, not dismissed. Establishing non-graphicness alone would still be too weak: one must consider logarithmic defect and both primal and dual alternatives.

There is a broader comparison. Goldberg–Jerrum approximate the positive random-cluster partition function on any regular matroid.[9] Conditioning one element to be absent or present leaves a deletion or contraction minor, which remains regular. Comparing the two approximated partition functions samples that element's conditional law. With relative error $O(\varepsilon/m)$ per step and amplified success probability, $O(m)$ such steps give a polynomial approximate sampler. A uniform binary-kernel sample then lifts the random-cluster sample to the code Gibbs law. Therefore failing the Code Swendsen–Wang criterion alone would not establish a new sampling regime.

6.2 A checked logical distinction: one encoded qubit does not certify easy matroid structure

Take r disjoint blocks of three qubits. On each block include all seven nonidentity Z stabilizers, and leave one additional qubit untouched. This is a bounded-degree, bounded-weight CSS family with one logical qubit. A block's check matrix has rows equal to all nonzero vectors of $\mathbb{F}_2^3$. Its relation space is the $[7, 4, 3]$ Hamming code, and its syndrome space is the $[7, 3, 4]$ simplex code.

Neither is a graph cycle space. The Hamming case would require a simple graph of incidence rank three with seven edges; the largest possible has six. The simplex case has only even nonzero weights and minimum weight four, so a representing graph would be simple and bipartite, of incidence rank four. Such a graph has at most six edges, again fewer than seven. The same bounds hold for disconnected graphs by summing component ranks.

If a graphic cycle supercode contains the direct sum of these block spaces, restricting it to each block must add at least one dimension. The restricted spaces have disjoint supports, so their added dimensions sum: $\Delta \geq r$ for both primal and dual comparisons. This does not classify BDL and is not a self-correcting memory. It proves that one logical qubit and bounded checks cannot replace the actual algebraic comparator test.

7 Resource conventions and the physical bill

Resource	Requirement and resulting scope
Access	Known stabilizer checks, a recorded random embedding, bounded-range bounded-support commuting interactions, their coupling scale, and a specified thermal bath. A two-body implementation or perturbative gadget is an additional construction, not supplied by naming the Hamiltonian.
Preparation	Product/reset ancillas, randomness, classical memory, a stabilizer encoder, and the thermal syndrome sampler. Sample before the unknown input arrives when possible. Fixed alphabet, degree and temperature constants may be extremely large or unfavorable.
Quantum control	Binary symplectic elimination compiles a known encoder with polynomial classical work and $O(N^2)$ Clifford gates. If the supplied local control network has $Q = \mathrm{poly}(N)$ vertices, routing each gate adds at most $O(Q)$ overhead. Include all routing qubits and gates. The model must permit switching/isolating the memory Hamiltonian and bath during writing/reading, or supply a bounded-error compensation scheme; a circuit run under uncontrolled thermal noise is not covered.
Input protection	An unknown bare input cannot be purified against an error already inflicted before encoding. Its entry error is a boundary resource. For a circuit of G noisy locations, the conservative condition $\sum_j \epsilon_j \le \delta_{\mathrm{control}}$ gives a sufficient diamond-norm control budget; uniform $\epsilon_j = O(\delta_{\mathrm{control}}/G)$ is inverse polynomial. Constant-rate fault-tolerant preparation needs a separate proof.
Holding noise	Use the specified Gibbs-stationary Markov generator, with local energy-resolved errors satisfying the protected-observable condition and bounded rates. The general stability input need not impose detailed balance, but the concrete model above does. Logical jumps, uncontrolled non-Markovian noise and bath-rate growth are not covered.
Uncertainty	Fixed positive coefficient variations are covered by (17). For their estimates, let $\lambda_a = \beta J_a$. Then $\mathrm{TV}(p_\lambda, p_{\widehat{\lambda}}) \le \frac{1}{4}\sum_a |\lambda_a - \widehat{\lambda}_a|$. Required coefficient estimates are an explicit access/calibration resource. Inverse-polynomial precision suffices for initialization; noncommuting fabrication errors and time drift are not covered.
Readout	Measure the stabilizer syndrome without learning the logical input, compute the same decoder correction, apply it, and invert the ground encoder. Include measurement, classical processing, routing and input/output errors.
Reuse	Each use needs fresh or reset ancillas and a fresh thermal syndrome sample. The stored qubit is not copied. Repeated uses accumulate error unless the stated accuracy budget is adjusted. Reset and refrigeration costs must be recorded.
Full holding cost	The physical system occupies space for time t. Its spacetime volume is at least proportional to Nt, and operating a refrigerator for that time has an explicit power-times-duration cost. Polynomial write/read cost does not imply polynomial total expenditure for an exponentially long hold.

The source constants are severe: its explicit random-embedding estimate uses a scale parameter $\lambda \approx 5.4 \times 10^{24}$ and density bound 6.6×10^{10}.[1] Calling these constants independent of N does not make them engineering-scale. Neither the unknown compiler constants nor an actual refrigerator/control energy budget have been numerically optimized here.

The coefficient-uncertainty estimate follows without a many-body norm estimate. Interpolate the dimensionless energies by $A_u(s) = A_0(s) + u\Delta A(s)$, so $\partial_u p_u(s) = -p_u(s)(\Delta A(s) - \mathbb{E}_u \Delta A)$. A variable in an interval of width R has mean absolute deviation at most $R/2$. Therefore $\frac{1}{2}\sum_s |\partial_u p_u(s)| \le R/4$, and here $R \le \sum_a |\lambda_a - \widehat{\lambda}_a|$. Integrating proves the stated bound. In combination with (17), this controls preparation and storage for the actual fixed positive-coefficient model. It does not supply a calibration experiment for a particular hardware platform; coefficient access to the stated precision

is part of the operation's input specification. Use conservative lower bounds satisfying the cooling condition for both actual and estimated coefficients.

General residual coherent Hamiltonian error is different. A direct perturbative channel bound can grow as $t\,\|\Delta H\|$; applying it at exponentially large t can demand exponentially small error. The present opening does not remove that obstruction. The defensible consequence is a foundational initialization theorem inside a specified commuting-Hamiltonian bath model, with an explicitly unfinished robustness problem.

8 What should receive sustained construction effort

The constructive object brought to the exchange is the source-conditional compiler and operation in Proposition 2, including both syndrome sectors and the separate protected decoder frame. Its proof is sufficiently explicit to be challenged on factor incidence, representative consistency and the physical control model. An implementation certificate would make those choices executable. Rare-cluster suppression alone would not have supplied the result.

The Code Swendsen–Wang and regular-matroid comparisons run alongside that construction. If an applicable representation exists, use it and revise the novelty claim immediately. If it does not, the history representation must still justify its own locality and contraction. Either outcome improves the problem selection rather than preserving a favored technique.

The conditional resolution combines three ingredients: (i) a thermal syndrome sampler at a fixed temperature compatible with memory stability; (ii) the exact reference-preserving loading reduction; and (iii) an explicit gate, routing, bath and uncertainty ledger. The broader reusable result is an algorithmic interface from contracting multiscale decoders to accessible thermal logical states. Its broad implication depends on which families satisfy the compiler hypotheses. Its status as a new discovery, rather than a useful corollary or reconstruction, requires the unresolved prior-method comparison to be settled.

8.1 Preserved results and independent selection

The completed equilibrium-inference theorem and quantum packages are unchanged. The equilibrium result remains a sharp law in its specified symmetric linear-response and sensor model, with its existing resource interpretation. The partial-SWAP acquisition-loss identity, compressor and accepted decoder bounds remain at their established scope; the unrestricted joint memory/processing law remains open. No claim in this brief upgrades those statements.

They were reusable alternatives, not obligations to remain in the same domain. Independent construction examined spin-glass sampling and thermal recovery as well as this memory route. The selected question was chosen for the concrete write/hold/read implication and the history compiler opening. No new tournament proposal from the other participant was used in making the choice.

Reproducibility and status

The companion package contains this source, the compiled report, exact example code and JSON outputs, independent challenge notes, and a source audit. Running `history_polymer_example.py` reproduces the finite algebraic checks. Those computations validate the toy representation only. They do not test a physical device, instantiate the complete BDL code, or independently validate its published lifetime proof.

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